

DATA SHEET

PCA1070

Multistandard programmable
analog CMOS transmission IC

Product specification
Supersedes data of 1996 Mar 06
File under Integrated Circuits, IC03

1997 Jun 20

Multistandard programmable analog CMOS transmission IC

PCA1070

FEATURES

- Line interface with:
 - Voltage regulator with programmable DC voltage drop
 - Programmable set impedance
 - Output to control an external switching MOS transistor for pulse dialling
 - Programmable DC voltage during pulse dialling
 - Circuitry for short DC settling time
- Interface to peripheral circuits with:
 - Supply for microcontroller and DTMF diallers
 - Input to sense supply voltage of microcontroller and output for reset of microcontroller
 - I²C-bus (programming of parameters, control of all logic signals)
 - High impedance DTMF signal input
 - Input for external oscillator signal with on-chip DC blocking
 - Power-down via the I²C-bus
 - Stabilized supply for electret microphone
- Microphone and DTMF amplifiers:
 - Low-noise microphone preamplifier suitable for various types of microphones
 - Symmetrical high impedance microphone preamplifier inputs
 - Programmable gain for microphone and DTMF channels
 - Sending mute via the I²C-bus to disable microphone amplifier and enable DTMF amplifier
 - Sending mute also to be used as privacy switch
 - Dynamic limiting (speech controlled) to prevent distortion of line signal and sidetone; programmable maximum sending level
- Receiving amplifier:
 - Suitable for various types of earpieces (including piezo)
 - Programmable gain and hearing protection level
 - Receiving mute via the I²C-bus to disable receiving amplifier and enable DTMF confidence tone
 - On-chip anti-sidetone circuit with programmable sidetone balance
 - Confidence tone in the earpiece during DTMF dialling



- Facility to regulate parameters with line current:
 - Value of DC line current (bit code) readable via the I²C-bus
 - Line loss compensation with fully software programmable characteristics (control range, stop current) of microphone/earpiece/DTMF amplifiers
 - Fully software programmable control of sidetone balance and DC voltage drop as a function of line length.

APPLICATIONS

- Wired telephony (basic till feature phones)
- Combi-terminals (e.g. telephone and answering machine or FAX)
- Modems and base units of cordless telephones.

GENERAL DESCRIPTION

The PCA1070 is a CMOS integrated circuit performing all speech and line interface functions in fully electronic telephone sets. The device requires a minimum of external components. The transmission parameters are programmable via the I²C-bus. This makes the IC adaptable to nearly all worldwide country requirements and to a various range of speech transducers, without changing the (few) external components.

The parameters are stored in the EEPROM of a microcontroller and are loaded into the PCA1070 during the start-up phase of the transmission IC after hook-off.

The PCA1070 also allows adaptation to the connected telephone line by reading the line current via the I²C-bus and processing it in the microcontroller.

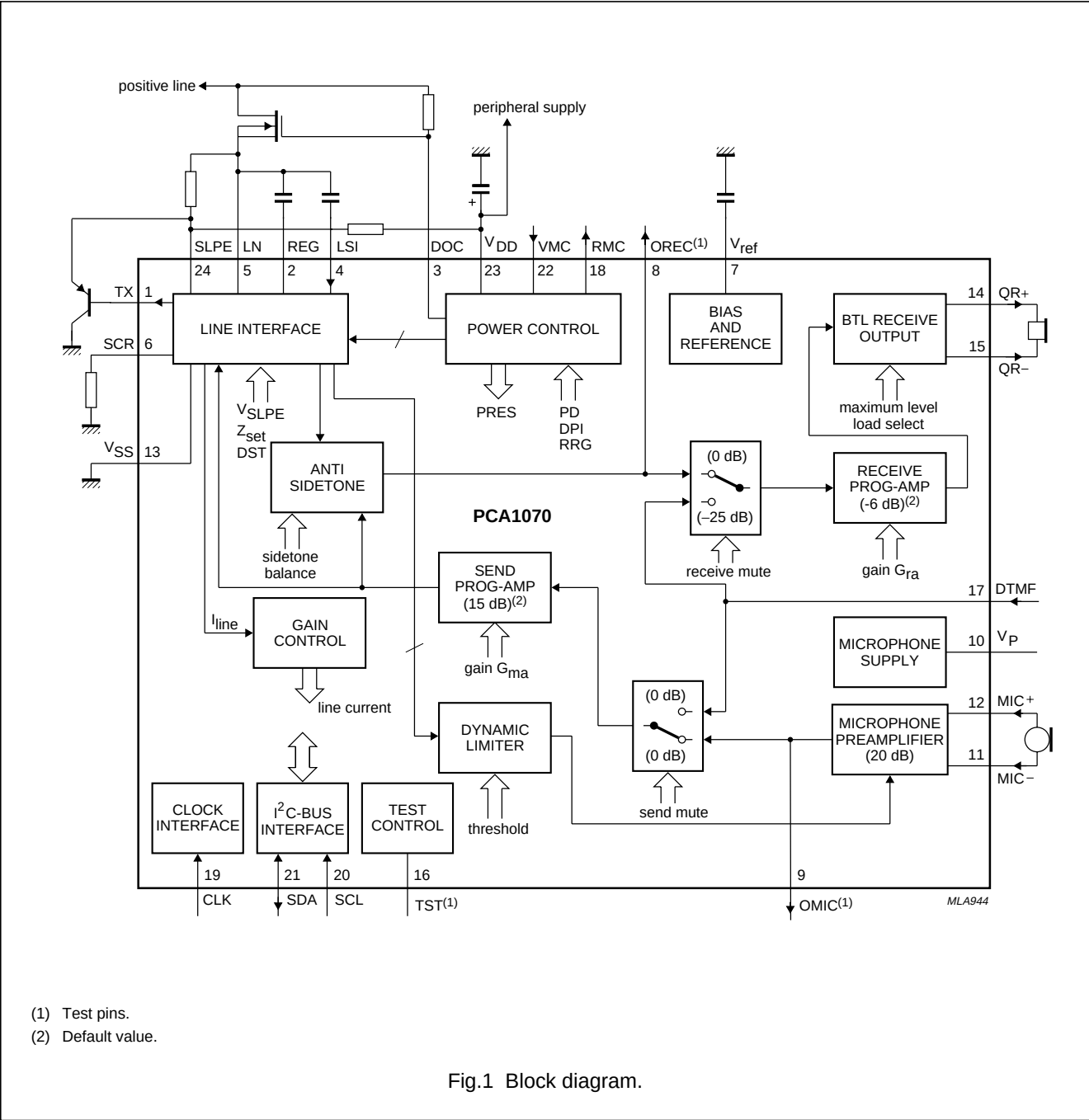
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ORDERING INFORMATION

TYPE NUMBER	PACKAGE		
	NAME	DESCRIPTION	VERSION
PCA1070P	DIP24	plastic dual in-line package; 24 leads (600 mil)	SOT101-1
PCA1070T	SO24	plastic small outline package; 24 leads; body width 7.5 mm	SOT137-1

BLOCK DIAGRAM



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PINNING

SYMBOL	PIN	DESCRIPTION
TX	1	drive output
REG	2	voltage regulator decoupling
DOC	3	dial output connection
LSI	4	line signal input
LN	5	positive line terminal
SCR	6	sending current resistor
V _{ref}	7	voltage reference decoupling
OREC	8	output receiving preamplifier; to be left open-circuit in application
OMIC	9	output microphone preamplifier; to be left open-circuit in application
V _P	10	supply for electret microphones
MIC ⁻	11	inverting input microphone preamplifier
MIC ⁺	12	non-inverting input microphone preamplifier
V _{SS}	13	negative line terminal
QR ⁺	14	non-inverting output of receiving amplifier
QR ⁻	15	inverting output of receiving amplifier
TST	16	test pin; to be connected to V _{SS} in application
DTMF	17	dual tone multi-frequency input
RMC	18	reset output for microcontroller
CLK	19	clock signal input
SCL	20	serial clock line input; I ² C-bus
SDA	21	serial data line input/output; I ² C-bus
VMC	22	input to sense supply voltage microcontroller
V _{DD}	23	positive supply decoupling
SLPE	24	slope (DC resistance) adjustment

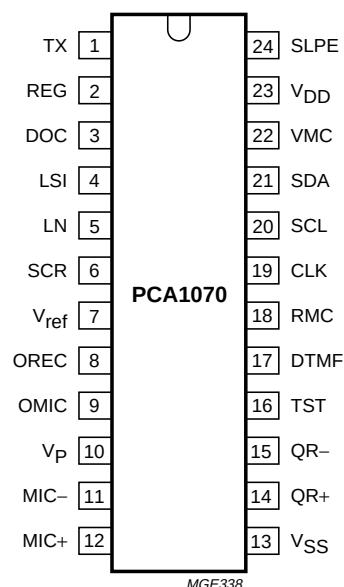


Fig.2 Pin configuration.

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FUNCTIONAL DESCRIPTION

All values in the Chapter "Functional description" are typical unless stated otherwise.

Line interface

DC VOLTAGE DROP

Power for the PCA1070 and its peripheral circuits is obtained from the telephone line. The IC develops its own supply voltage at V_{DD} and regulates its DC voltage drop between pins SLPE and V_{SS} . This voltage (V_{SLPE}) can be programmed via the I²C-bus interface between 3.1 to 5.9 V and is default at 4.7 V (see Table 8).

The DC line voltage at pin LN can be calculated using the following equation:

$$V_{LN} = V_{SLPE} + (I_{line} - I_{LN}) \times R_{LN-SLPE}$$

where:

I_{LN} = DC bias current flowing into pin LN
(≈ 3 mA if $I_{line} > 17$ mA)

$R_{LN-SLPE}$ = external 20 Ω resistor between LN and SLPE.

At line currents below 6 mA the DC voltage V_{SLPE} is automatically adjusted to a lower value. This means that the operation of more sets, connected in parallel, is possible with reduced sending and receiving levels and relaxed performance. At line currents below 16 mA the DC bias current I_{LN} is reduced from ≈ 3 mA to a lower value to ensure maximum possible transmit level capability under all line current conditions.

SET IMPEDANCE

In normal conditions $I_{line} \gg I_{LN}$ and the static behaviour is equivalent to a voltage regulator diode with a series resistor $R_{LN-SLPE}$. In the audio frequency range the dynamic impedance Z_{LN} is determined mainly by the internal component $Z_{set} = R_a + (R_b \parallel C)$. The equivalent impedance Z_{LN} is shown in Fig.3. The values of R_a , R_b and C can be programmed via the I²C-bus interface (see Tables 9, 10 and 11).

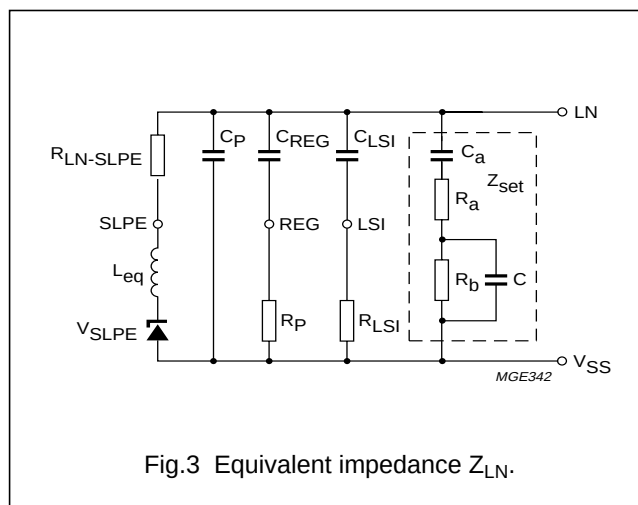


Fig.3 Equivalent impedance Z_{LN} .

where:

C_a = DC blocking capacitor (influence negligible at $f \geq 300$ Hz for given value of C_{LSI})

C_{LSI} = capacitor at pin LSI (100 nF)

C_P = internal capacitor (12 nF)

C_{REG} = capacitor at pin REG (470 nF)

L_{eq} = artificial inductor
($= R_P \times R_{LN-SLPE} \times C_{REG} = 10.1$ H at $V_{SLPE} = 4.7$ V)

$R_{LN-SLPE}$ = DC slope resistance (20 Ω)

R_P = internal resistor (1075 k Ω at $V_{SLPE} = 4.7$ V)

R_{LSI} = internal resistor (240 k Ω).

SUPPLY FOR PERIPHERAL CIRCUITS

The supply voltage V_{DD} can be used for peripheral circuitry. The supply capabilities depend on the programmed DC voltage drop V_{SLPE} and on several other parameters as given in the following equation:

$$V_{DD} = V_{SLPE} - (I_{DD} + I_p + I_{VP}) \times R_{SLPE-VDD}$$

where:

I_{DD} = internal current consumption PCA1070 (2.3 mA)

I_p = current to peripheral circuitry

I_{VP} = current taken from V_P for electret microphone

$R_{SLPE-VDD}$ = external resistor between SLPE and V_{DD} .

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DC STARTING AND SETTLING TIME

The IC is equipped with circuitry for fast DC start-up. This circuit is automatically activated as soon as V_{DD} reaches 3 V after hook-off, and is deactivated when V_{SLPE} drops below 5.9 V. This ensures that only a relatively short time is needed to reach the default DC setting (V_{SLPE}) of the circuit and that V_{DD} will not exceed the maximum permitted voltage of 6 V.

The start-up circuit can also be activated under software control by setting bit code DST to logic 1 via the I²C-bus. The start-up time can be optimized by programming the bit code DST to logic 1 during the start-up procedure. In practice this is possible as soon as the microcontroller has become operational. The DST bit can also be used to quickly restore the DC settings (V_{SLPE}) after long line breaks or during reprogramming of V_{SLPE} .

It should be noted that the AC impedance into pin LN is reduced considerably when DST = 1.

Power control

INTERNAL RESET PCA1070

The PCA1070 has an internal reset circuit that monitors the supply voltage V_{DD} . If V_{DD} is below the threshold level (1.2 V) then the circuit is in reset-mode. In this mode the current consumption is low and the internal reset is active and writes the default values into all registers. The status bit PRES will be set to logic 1. The microcontroller can read this bit via the I²C-bus interface; once read it will be set to logic 0 again.

When V_{DD} passes the threshold (increasing V_{DD}), the circuit becomes partly active and the internal ring/speech detector will be activated (see Section "Start-up and switch-off behaviour").

RESET OUTPUT FOR MICROCONTROLLER

The voltage at pin VMC (microcontroller supply voltage) is monitored by a reset circuit. If V_{VMC} is below the threshold level the output RMC is set to logic 1. This threshold level is 2 V in the normal operating and power-down mode and 2.1 V in the standby mode (see Fig.4).

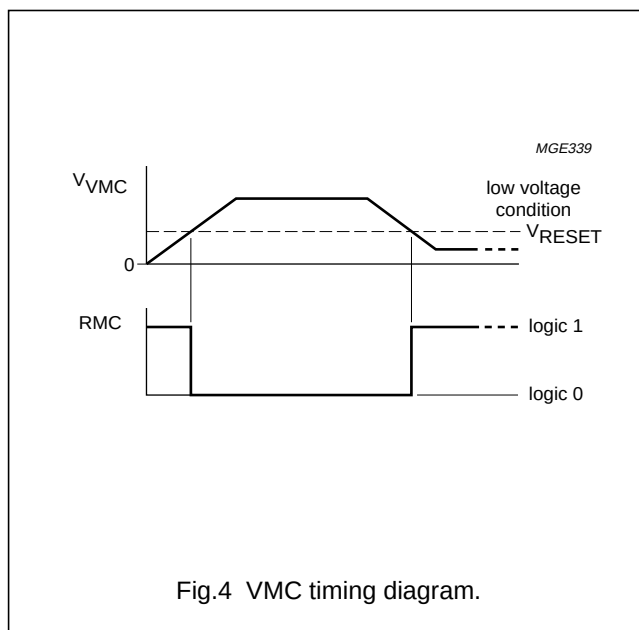


Fig.4 VMC timing diagram.

POWER-DOWN/STANDBY MODES

The circuit can be set in power-down or standby mode. These modes are intended for use with pulse dialling during long line breaks and applications with memory retention.

With control bits PDx = 01, the circuit is in the power-down mode; the typical current consumption at pin V_{DD} is reduced from $I_{DD} = 2.3$ mA to 30 μ A; the typical current consumption at pin VMC is 4 μ A. When PDx = 11 the circuit is in the standby mode and I_{DD} and I_{VMC} are reduced to 2 μ A. In both conditions (power-down and standby) the voltage stabilizer will be disabled.

START-UP AND SWITCH-OFF BEHAVIOUR

This description refers to the basic application where V_{DD} and VMC are connected together and one supply capacitor is used (see Fig.8).

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Speech condition

After hook-off, line current will be applied to the line input LN and the supply capacitor connected to V_{DD} and VMC will be charged.

The internal reset signal will change from logic 1 to logic 0 when V_{DD} passes the threshold level (1.2 V) and the circuit becomes partly active [the line interface part is kept in power-down mode, so that all of the line current is available to charge the supply capacitor(s)];

The PCA1070 can receive data via the I²C-bus (standard I²C specifications are fulfilled for $V_{DD} \geq 2.5$ V; relaxed performance for $V_{DD} = 1.8$ to 2.5 V).

When V_{VMC} passes the microcontroller reset level of 2 V (2.1 V in standby mode) the output RMC changes from logic 1 to logic 0 and the circuit is switched to the normal operating mode.

After hook-on V_{VMC} decreases and the output RMC will change from logic 0 to logic 1 when V_{VMC} passes the threshold level, however the PCA1070 will stay in the normal operating mode until the internal reset at 1.2 V takes place.

By decreasing V_{DD} the internal reset signal will change from logic 0 to logic 1 when V_{DD} passes 1.2 V and the circuit will go into the reset mode (line interface part in power-down and all programmable parameters reset to default values).

Ringer condition

In this condition the supply capacitor connected to V_{DD} and VMC is charged by the rectified ringer signal; no line current is applied to pin LN.

V_{DD} and V_{VMC} are increasing and when V_{DD} passes the internal reset threshold level (1.2 V), the internal ring/speech-detector will be activated and the circuit will switch to the standby condition ($I_{DD} < 5 \mu\text{A}$; $I_{VMC} < 5 \mu\text{A}$) before the voltage at VMC reaches the threshold level for microcontroller reset. When V_{VMC} passes this threshold level (2.1 V) output RMC changes from logic 1 to logic 0 and the circuit will stay in the standby mode until line current is applied to pin LN. By setting the 'Reset Ring' control bit (RRG) to logic 1 via the I²C-bus interface, the ring/speech detector will be disabled.

DIAL PULSE INPUT (DPI)

The DPI bit controls output DOC (open-drain) that drives the gate of an external MOS interrupter transistor. DPI is controlled via the I²C-bus interface.

If DPI is set to logic 1, pin DOC will be pulled down to switch-off the MOSFET to generate a line break. If DPI = 0 pin DOC is high-ohmic and the interrupter transistor will conduct the line current.

Sending channel

The PCA1070 has symmetrical microphone inputs and accepts input signals of maximum 70 mV (peak) for THD = 2% ($V_{DD} \geq 2.5$ V). Its input impedance is 100 k Ω and its gain is default 41 dB. Dynamic, magnetic, piezoelectric and electret (with built-in FET source follower) microphones can be used. Some possible microphone arrangements are shown in Fig.5.

The gain of the sending channel can be programmed between 30 dB and 51 dB in 1 dB steps using bit code GMAx (6 bits). The gain of the microphone preamplifier is 20 dB (with dynamic limiter not active) and GMAx sets the gain of the 'sending prog-amp' (allowed range $G_{ma} = 4$ to 25 dB). The gain of the line interface is 6 dB.

Thus the total gain of the sending channel (G_M) is as follows:

$$G_M = 20 + G_{ma} + 6 \text{ (dB)}$$

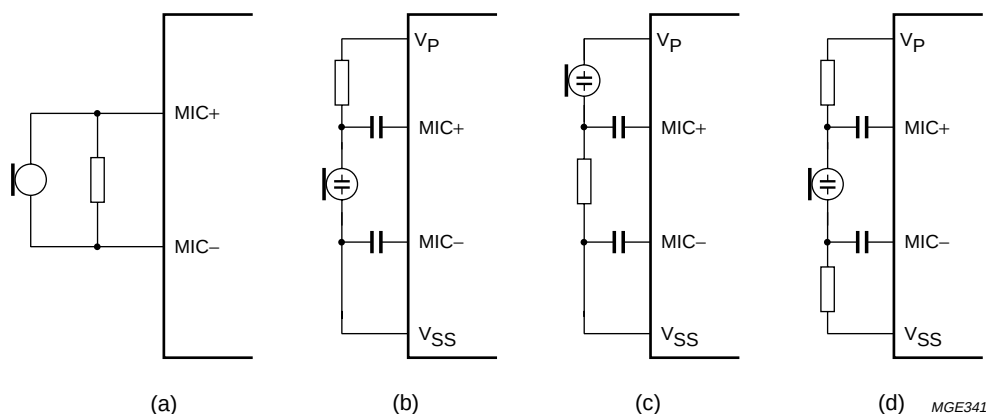
$$\text{Default: } G_M = 20 + 15 + 6 = 41 \text{ dB}$$

Where G_{ma} = 'gain sending prog-amp'.

Programming the gain of the 'sending prog-amp' is given in Table 13.

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- (a) Dynamic or piezo.
 (b) Low impedance electret with built-in pre-amplifier.
 (c) High impedance electret with built-in pre-amplifier.
 (d) Symmetrical connection of electret.

Fig.5 Microphone arrangements.

Dynamic limiter

To prevent distortion of the transmitted speech signal, the gain of the microphone amplifier is reduced rapidly when signal peaks on the line exceed an internally determined threshold level. The time in which the gain is reduced, the attack time, is very short. The circuit stays in this gain-reduced condition until the peaks of the sending signal remain below the threshold level. The sending gain then returns to normal after a time also determined on the chip, the release time. The threshold level of the AC peak-to-peak line voltage on pin LN is default at 3.5 V (p-p). A level of 2.6 V (p-p) can be programmed by setting bit code DLT to logic 1.

The internal threshold level is lowered automatically if the DC voltage setting of the circuit (V_{SLPE}) is not high enough to reach the programmed level. Also when the DC current in the transmit output stage is insufficient to drive the line load, the internal threshold level is lowered automatically.

Dynamic limiting considerably improves sidetone performance in over-drive conditions (less distortion and limited sidetone level).

DTMF channel

The PCA1070 has an asymmetrical DTMF input. Its input impedance is $200\text{ k}\Omega // 45\text{ pF}$ and its gain is default at 21 dB. DTMF signals can be sent to the line by setting control bit 'Sending Mute' (SM) to logic 1 (default SM = 0); by setting 'Receiving Mute' (RM) also to logic 1 (default RM = 0), the dialling tones are also sent to the receiving output to generate a confidence tone in the earpiece.

The gain between the DTMF input and the line LN can be programmed between 1 dB and 21 dB in 1 dB steps using bit code GMAx (6 bits). The confidence tone gain (between DTMF input and earpiece outputs QR) can be programmed between -40 dB and -19 dB (symmetrical drive of earpiece) using bit code GRAx (6 bits). GMAx sets the gain of the 'sending prog-amp' (recommended range in DTMF mode for $G_{ma} = -5$ to 15 dB) and GRAx sets the gain of the 'rec prog-amp' (allowed range $G_{ra} = -25$ to 0 dB).

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The total gain of the DTMF channel between the DTMF input and the line LN is as follows:

$$G_{\text{DTMF}} = G_{\text{ma}} + 6 \text{ (dB)}$$

$$\text{Default } G_{\text{DTMF}} = 15 + 6 = 21 \text{ dB}$$

The confidence tone gain (DTMF to QR outputs) is:

With symmetrical drive of earpiece $G_{\text{CTS}} = G_{\text{ra}} - 19 \text{ (dB)}$

$$\text{Default } G_{\text{CTS}} = -6 - 19 = -25 \text{ dB.}$$

At low gain settings ($G_{\text{ra}} < -10 \text{ dB}$), the confidence tone gain will be slightly higher than the calculated value. This is caused by a residual signal.

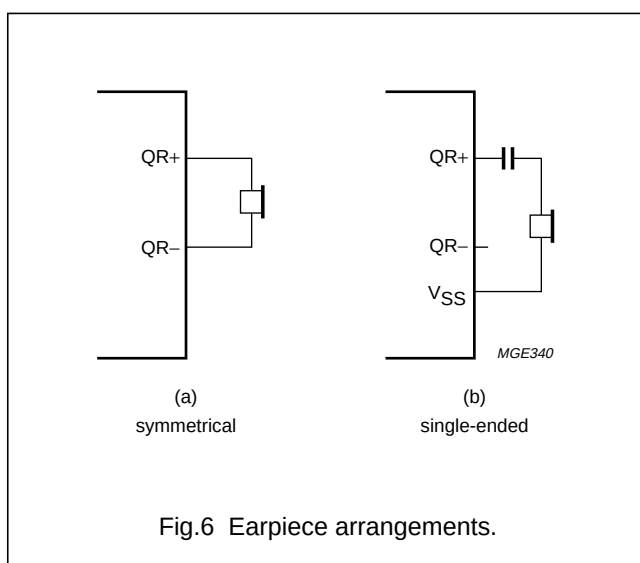
Programming the gain of the 'sending prog-amp' and the 'rec prog-amp' is given in Table 13.

Receiving channel

The gain of the receiving channel is defined between the line connection LN and the earpiece outputs QR+ and QR-. Its voltage gain is default -6 dB (differential drive). The LN terminal accepts receiving signals up to 1 V (RMS) for THD = 2%. The outputs may be used to connect dynamic, magnetic or piezoelectric earpieces with single-ended or differential drive. The load select bit RFC is set default to logic 1 to guarantee stable operation in case of a capacitive load (piezoelectric earpiece). With a resistive load (dynamic capsule) RFC should be set to logic 0 via the I²C-bus interface to obtain optimum performance with respect to distortion and bandwidth.

Two levels for hearing protection can be selected via the I²C-bus interface with control bit HPL.

The earpiece arrangements are illustrated in Fig.6.



The gain of the receiving channel can be programmed between -19 dB and +11 dB (symmetrical drive) in 1 dB steps using bit code GRAX (6 bits).

GRAX sets the gain of the 'rec prog-amp' (allowed range $G_{\text{ra}} = -19 \text{ dB}$ to $+11 \text{ dB}$; default $G_{\text{ra}} = -6 \text{ dB}$).

The total gain of the receiving channel is as follows:

$$\text{Symmetrical drive } G_{\text{RS}} = G_{\text{ra}} \text{ (dB)}$$

$$\text{Default } G_{\text{RS}} = -6 \text{ dB.}$$

Asymmetrical or single-ended drive $G_{\text{RA}} = G_{\text{RS}} - 6 \text{ (dB)}$

$$\text{Default } G_{\text{ra}} = -6 - 6 \text{ (dB)} = -12 \text{ dB.}$$

Programming the gain G_{ra} of the 'rec prog-amp' is given in Table 13.

Sidetone balance

The PCA1070 has an on-chip anti-sidetone circuit. An internal balance impedance Z_{oss} can be programmed via the I²C-bus interface to match the external line impedance Z_{line} to give optimum sidetone suppression.

$$Z_{\text{oss}} = R_{\text{sa}} + (R_{\text{sb}} // C_{\text{s}}).$$

Programming the sidetone balance impedance is given in Tables 14, 15 and 16.

Line current control

The DC line current can be read via the I²C-bus interface. This information can be used for the adaptation of transmission parameters (for example line loss compensation, sidetone balance and DC characteristic).

The bit code LCx as a function of line current is given in Table 17.

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I²C-BUS PROGRAMMING

Table 1 Programmable parameters

The following parameters (see Fig.1) can be programmed by means of a bit code via the I²C-bus:

SYMBOL	PARAMETER	BLOCK	BITS	DESCRIPTION
VDCx	V _{SLPE}	line interface	3	DC voltage SLPE-V _{SS}
ZSAX	set impedance	line interface	3	R _a of set impedance
ZSBx		line interface	3	R _b of set impedance
ZSPx		line interface	4	f _p (pole frequency) of set impedance
DST	DST	line interface	1	DC Start Time
PDx	PD	power control	2	Power-Down
DPI	DPI	power control	1	Dial Pulse Input
RRG	RRG	power control	1	Reset RinG detector
HPL	maximum receiving level	BTL receiving output	1	Hearing Protection Level
RFC	load select	BTL receiving output	1	Resistive/Capacitive load
ZOSAx	sidetone impedance	anti-sidetone	4	R _{sa} of sidetone impedance
ZOSBx		anti-sidetone	4	R _{sb} of sidetone impedance
ZOSPx		anti-sidetone	4	C _s of sidetone impedance
RM	receiving mute	receiving mute	1	Receiving Mute
GRAX	gain G _{ra}	receiving prog-amp	6	Gain receiving prog-amp
GMAX	gain G _{ma}	sending prog-amp	6	Gain sending prog-amp
SM	sending mute	sending mute	1	Sending Mute
DLT	threshold	dynamic limiter	1	Dynamic Limiter Threshold

Table 2 Readable parameters

The following parameters (see also Fig.1) can be read as a bit code via the I²C-bus:

SYMBOL	PARAMETER	BLOCK	BITS	DESCRIPTION
PRES	PRES	power control	1	PCA1070 Reset
LCx	line current	gain control	5	Line Current

I²C interface

The I²C-bus interface (see “The I²C-bus and how to use it” 12NC: 9398 393 40011) is used to program the transmission parameters and control functions.

Table 3 Device address

A6	A5	A4	A3	A2	A1	A0	R/W
0	1	0	0	0	1	0	X

All functions can be accessed by writing an 8-bit word to the PCA1070. In order to set up the PCA1070, a control message consisting of the device address, a R/W bit, a subaddress byte and one or more data bytes must be written to the PCA1070. If more than one data byte follows the subaddress, these bytes are stored in the successive registers by the automatic increment feature.

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Table 4 The control word format for the slave receiver

DEVICE ADDRESS										SUB ADDRESS										DATA/CONTROL BYTE									
S	0	1	0	0	0	1	0	0	0 ⁽¹⁾	A	I7	I6	I5	I4	I3	I2	I1	I0	A	D7	D6	D5	D4	D3	D2	D1	D0	A	P

Note

1. This bit is $\overline{R/\overline{W}}$.

Table 5 Bit arrangement of each data byte used in the control word: PCA1070 receive (see note 1)

FUNCTION	SUB ADDRESS	D7	D6	D5	D4	D3	D2	D1	D0
DC voltage	H00		VDC2	VDC1	VDC0				DST
Sidetone and set impedance	H01	ZOSB3	ZOSB2	ZOSB1	ZOSB0	ZOSA3	ZOSA2	ZOSA1	ZOSA0
	H02	ZOSP3	ZOSP2	ZOSP1	ZOSP0		ZSA2	ZSA1	ZSA0
	H03		ZSB2	ZSB1	ZSB0	ZSP3	ZSP2	ZSP1	ZSP0
Sending channel	H04	DLT		GMA5	GMA4	GMA3	GMA2	GMA1	GMA0
Receiving channel	H05	RFC	HPL	GRA5	GRA4	GRA3	GRA2	GRA1	GRA0
Control	H06	PD1	PD0		RRG	RM	SM		DPI

Note

1. The bits that are not indicated must be set to logic 0.

Table 6 The control word format for the slave transmitter

DEVICE ADDRESS										DATA/STATUS BYTE									
S	0	1	0	0	0	1	0	1 ⁽¹⁾	A	D7	D6	D5	D4	D3	D2	D1	D0	A	P

Note

1. Change in direction of $\overline{R/\overline{W}}$ bit.

Table 7 PCA1070 send

FUNCTION	D7	D6	D5	D4	D3	D2	D1	D0
PCA1070 status	PRES ⁽¹⁾	—	—	LC4 ⁽²⁾	LC3 ⁽²⁾	LC2 ⁽²⁾	LC1 ⁽²⁾	LC0 ⁽²⁾

Notes

1. Indicates if PCA1070 has received internal reset; PRES will be set to logic 1 with internal reset and is set to logic 0 after reading the register via the I²C-bus.
2. Information about value of line current.

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WRITE AND READ TABLES

DC voltages

Table 8 DC voltage at pin SLPE

VDC2	VDC1	VDC0	V _{SLPE} (V)	REMARK
0	0	0	3.1	
0	0	1	3.5	
0	1	0	3.9	
0	1	1	4.3	
1	0	0	4.7	default
1	0	1	5.1	
1	1	0	5.5	
1	1	1	5.9	

Set impedance

Programming the impedance in the audio frequency range seen at pin LN: $R_a + (R_b // C)$

Table 9 Programming R_a

ZSA2	ZSA1	ZSA0	R_a (Ω)	REMARK
0	0	0	0	
0	0	1	100	
0	1	0	200	default
0	1	1	300	
1	0	0	400	
1	0	1	500	note 1
1	1	X	600	notes 1 and 2

Notes

- For Z_{set} combinations where $R_a = 0$ only $R_b = 600 \Omega$ is allowed. If $R_a \geq 500 \Omega$ it is obligatory that $R_b = 0$. This is to safeguard stable operation of the line interface under all practical conditions. If Z_{ref} requires $R_a = 0$ and $R_b \neq 600 \Omega$ use $R_a = 100 \Omega$ instead and reduce the original R_b by 100Ω .
- X = don't care.

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Table 10 Programming R_b

ZSB2	ZSB1	ZSB0	R_b (Ω)	REMARK
0	0	0	0	note 1
0	0	1	600	
0	1	0	700	
0	1	1	800	default
1	X	0	900	note 2
1	X	1	1000	note 2

Notes

- For Z_{set} combinations where $R_a = 0$ only $R_b = 600 \Omega$ is allowed. If $R_a \geq 500 \Omega$ it is obligatory that $R_b = 0$. This is to safeguard stable operation of the line interface under all practical conditions. If Z_{ref} requires $R_a = 0$ and $R_b \neq 600 \Omega$ use $R_a = 100 \Omega$ instead and reduce the original R_b by 100Ω .
- X = don't care.

Table 11 Programming pole frequency:

ZSP3	ZSP2	ZSP1	ZSP0	f_p (Hz)	CORRESPONDING VALUE OF C (nF) ⁽¹⁾					REMARK
					R_b (600 Ω)	R_b (700 Ω)	R_b (800 Ω)	R_b (900 Ω)	R_b (1000 Ω)	
0	0	0	0	828	320	275	240	214	192	
0	0	0	1	1095	242	207	182	161	145	
0	0	1	0	1448	183	157	137	122	110	
0	0	1	1	1915	139	119	104	92	83	default
0	1	0	0	2533	105	90	79	70	63	
0	1	0	1	3350	79	68	59	53	48	
0	1	1	0	4430	60	51	45	40	36	
0	1	1	1	5859	45	39	34	30	27	
1	X	X	X	12000	22	19	17	15	13	note 2

Notes

- $C = \frac{1}{2\pi \times R_b \times f_p}$
- X = don't care.

Reset functions

Monitoring of internal reset PCA1070.

Table 12 Status bit PRES

PRES	DESCRIPTION
1	internal reset has occurred; default values in all registers
0	register has been read via the I ² C-bus interface

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Programmable amplifier (prog-amp)

An identical programmable amplifier called 'prog-amp' is used both in the sending and receiving channel. The bit codes GMAx and GRAx are given in Table 13. The permitted adjustment range differs for the two amplifiers and is also different for DTMF and speech mode. This is indicated in the corresponding sections.

Table 13 Bit code prog-amp

GAIN (dB)	GMA5	GMA4	GMA3	GMA2	GMA1	GMA0
	GRA5	GRA4	GRA3	GRA2	GRA1	GRA0
-25	1	1	1	0	0	1
-24	1	1	1	0	0	0
-23	1	1	0	1	1	1
-22	1	1	0	1	1	0
-21	1	1	0	1	0	1
-20	1	1	0	1	0	0
-19	1	1	0	0	1	1
-18	1	1	0	0	1	0
-17	1	1	0	0	0	1
-16	1	1	0	0	0	0
-15	1	0	1	1	1	1
-14	1	0	1	1	1	0
-13	1	0	1	1	0	1
-12	1	0	1	1	0	0
-11	1	0	1	0	1	1
-10	1	0	1	0	1	0
-9	1	0	1	0	0	1
-8	1	0	1	0	0	0
-7	1	0	0	1	1	1
-6 ⁽¹⁾	1	0	0	1	1	0
-5	1	0	0	1	0	1
-4	1	0	0	1	0	0
-3	1	0	0	0	1	1
-2	1	0	0	0	1	0
-1	1	0	0	0	0	1
0	1	0	0	0	0	0

GAIN (dB)	GMA5	GMA4	GMA3	GMA2	GMA1	GMA0
	GRA5	GRA4	GRA3	GRA2	GRA1	GRA0
+0	0	0	0	0	0	0
+1	0	0	0	0	0	1
+2	0	0	0	0	1	0
+3	0	0	0	0	1	1
+4	0	0	0	1	0	0
+5	0	0	0	1	0	1
+6	0	0	0	1	1	0
+7	0	0	0	1	1	1
+8	0	0	1	0	0	0
+9	0	0	1	0	0	1
+10	0	0	1	0	1	0
+11	0	0	1	0	1	1
+12	0	0	1	1	0	0
+13	0	0	1	1	0	1
+14	0	0	1	1	1	0
+15 ⁽²⁾	0	0	1	1	1	1
+16	0	1	0	0	0	0
+17	0	1	0	0	0	1
+18	0	1	0	0	1	0
+19	0	1	0	0	1	1
+20	0	1	0	1	0	0
+21	0	1	0	1	0	1
+22	0	1	0	1	1	0
+23	0	1	0	1	1	1
+24	0	1	1	0	0	0
+25	0	1	1	0	0	1

Notes

1. Default value 'rec prog-amp' GRAx.
2. Default value 'sending prog-amp' GMAx.

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Sidetone balance impedance

Internal balance impedance Z_{oss} to match the external line impedance Z_{line} to give optimum sidetone suppression.

$$Z_{oss} = R_{sa} + (R_{sb} // C_s).$$

The optimum setting of R_{sa} depends on the value of the set impedance. To safeguard stable operation of the anti-sidetone circuit under all practical conditions, the following condition must be fulfilled: $R_{sa} \geq 0.5R_a$.

Table 14 Programming R_{sa}

ZOSA3	ZOSA2	ZOSA1	ZOSA0	$R_{sa} (\Omega)$
0	0	0	0	134
0	0	0	1	153
0	0	1	0	193
0	0	1	1	221
0	1	0	0	246
0	1	0	1	277
0	1	1	0	295
0	1	1	1	341
1	0	0	0	369
1	0	0	1	443
1	0	1	0	492⁽¹⁾
1	0	1	1	–
1	1	0	0	–
1	1	0	1	–
1	1	1	0	–
1	1	1	1	–

Note

1. Default value.

Table 15 Programming R_{sb}

ZOSB				$R_{sb} (\Omega)$
MSB			LSB	
0	0	0	0	465
0	0	0	1	637
0	0	1	0	710
0	0	1	1	803
0	1	0	0	893
0	1	0	1	1003
0	1	1	0	1259⁽¹⁾
0	1	1	1	1410
1	0	0	0	1572
1	0	0	1	1773
1	0	1	0	1978
1	0	1	1	2216
1	1	0	0	–
1	1	0	1	–
1	1	1	0	–
1	1	1	1	–

Note

1. Default value.

Table 16 Programming C_s

ZOSP				$C_s (nf)$
MSP			LSP	
0	0	0	0	5
0	0	0	1	55
0	0	1	0	58
0	0	1	1	69
0	1	0	0	76
0	1	0	1	85
0	1	1	0	96
0	1	1	1	105
1	0	0	0	121
1	0	0	1	134⁽¹⁾
1	0	1	0	145
1	0	1	1	166
1	1	0	0	186
1	1	0	1	207
1	1	1	0	232
1	1	1	1	259

Note

1. Default value.

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Line current control

Table 17 Bit code LCx and DC line current

LC4	LC3	LC2	LC1	LC0	I _{line} (typ.) (mA)
0	0	0	0	0	<12.5
0	0	0	0	1	15.0
0	0	0	1	0	17.5
0	0	0	1	1	20.0
0	0	1	0	0	22.5
0	0	1	0	1	25.0
0	0	1	1	0	27.5
0	0	1	1	1	30.0
0	1	0	0	0	32.5
0	1	0	0	1	35.0
0	1	0	1	0	37.5
0	1	0	1	1	40.0
0	1	1	0	0	42.5
0	1	1	0	1	45.0
0	1	1	1	0	47.5
0	1	1	1	1	50.0
1	0	0	0	0	52.5
1	0	0	0	1	55.0
1	0	0	1	0	58.0
1	0	0	1	1	61.0
1	0	1	0	0	64.0
1	0	1	0	1	66.5
1	0	1	1	0	69.0
1	0	1	1	1	71.5
1	1	0	0	0	74.0
1	1	0	0	1	77.5
1	1	0	1	0	80.0
1	1	0	1	1	82.5
1	1	1	0	0	85.0
1	1	1	0	1	88.0
1	1	1	1	0	91.0
1	1	1	1	1	>94.0

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LIMITING VALUES

In accordance with the Absolute Maximum System (IEC 134).

SYMBOL	PARAMETER	MIN.	MAX.	UNIT
V_{LN}	positive line voltage at pin LN	-0.8	+12	V
V_i	input voltage on pins SLPE, DOC, REG, TX and LSI	-0.8	+12	V
V_{DD}	supply voltage	-0.8	+7.0	V
V_n	voltage on all other pins	-0.8	+7.0	V
I_i	input current	-10	+10	mA
P_{tot}	total power dissipation	–	250	mW
T_{stg}	storage temperature	-40	+125	°C
T_{amb}	operating ambient temperature	-10	+60	°C

HANDLING

Inputs and outputs are protected against electrostatic discharge in normal handling. However, to be totally safe, it is desirable to take normal precautions appropriate to handling MOS devices.

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	VALUE	UNIT
$R_{th\ j-a}$	thermal resistance from junction to ambient in free air		
	DIP24	54	K/W
	SO24	74	K/W

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TEST CONDITIONS AND PARAMETER SETTINGS FOR THE CHARACTERISTICS

Table 18 Test conditions

SYMBOL	VALUE	UNIT
I_{line}	20	mA
V_{SS}	0	V
f	1000	Hz
I_p	0	A
I_{VP}	0	A
f_{clk}	3.597545	MHz
T_{amb}	25	°C
Z_{line}	220 Ω + 820 Ω // 115 nF	
R_m	150	Ω
R_t	150	Ω

Table 19 Test settings and control bits. All values, except RFC, are default. Programmable via the I²C-bus; bit codes are given in Chapter "I²C-bus programming".

SYMBOL	VALUE
VDCx	100
ZSAx	010
ZSBx	011
ZSPx	0011
GMAx	001111
GRAx	100110
ZOSAx	1010
ZOSBx	0110
ZOSPx	1001
DST	0
DLT	0
RFC	0
HPL	0
PDx	00
RRG	0
RM	0
SM	0
DPI	0

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CHARACTERISTICS

All parameters are measured in the test circuit of Fig.7 under the conditions specified in Tables 18 and 19; unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
DC line interface: LN, TX, SLPE and REG						
I_{line}	line current operating range		17	–	140	mA
		reduced sending level	12	–	17	mA
V_{SLPE}	DC voltage at SLPE	with or without clock	4.3	4.7	5.1	V
$V_{SLPE(min)}$	minimum selectable value	$V_{DCx} = 000$	2.8	3.1	3.4	V
$V_{SLPE(max)}$	maximum selectable value	$V_{DCx} = 111$	5.4	5.9	6.4	V
$V_{SLPE(step)}$	step resolution		–	0.4	–	V
V_{SLPE}	DC voltage at SLPE	with or without clock; fast start-up; DST = 1	–	4.7	–	V
$V_{SLPE(min)}$	minimum selectable value	fast start-up; DST = 1; $V_{DCx} = 000$	–	3.1	–	V
$V_{SLPE(max)}$	maximum selectable value	fast start-up; DST = 1; $V_{DCx} = 111$	–	5.9	–	V
$V_{SLPE(step)}$	step resolution	fast start-up; DST = 1	–	0.4	–	V
ΔV_{SLPE}	variation with temperature	at $T_{amb} = -10\text{ }^{\circ}\text{C}$ to $+60\text{ }^{\circ}\text{C}$ with respect to $25\text{ }^{\circ}\text{C}$	–	± 20	–	mV
V_{LN}	DC line voltage at LN	with or without clock	4.6	5.0	5.4	V
		$I_{line} = 12\text{ mA}$		4.83		V
		$I_{line} = 120\text{ mA}$	6.5	7.0	7.5	V
V_{LN}	DC line voltage at LN at low line current	with or without clock; $I_{line} = 0.25\text{ mA}$	–	1	–	V
		$I_{line} = 2\text{ mA}$	–	1.9	–	V
		$I_{line} = 4\text{ mA}$	–	3.4	–	V
		$I_{line} = 7\text{ mA}$	–	4.73	5.2	V
t_{DC}	DC start-up time	$C_{VDD} = 470\text{ }\mu\text{F}$; no clock; note 1	–	145	–	ms
TX: DRIVE OUTPUT FOR EXTERNAL PNP						
V_{TX}	output voltage at TX	external PNP disconnected; $V_{SLPE} = 2\text{ V}$; $V_{REG} = 1.5\text{ V}$; $V_{DD} = V_{VMC} = 2.5\text{ V}$; $I_{TX} = 0\text{ mA}$	–	1.45	–	V
		$V_{SLPE} = 3\text{ V}$; $V_{REG} = 2.5\text{ V}$; $V_{DD} = V_{VMC} = 2.5\text{ V}$; $I_{TX} = 1.6\text{ mA}$	–	2.2	–	V
t_{sw}	switching time DC voltage at SLPE	V_{SLPE} steps from 3.1 V to 5.9 V; note 2	–	65	–	ms
		V_{SLPE} steps from 5.9 V to 3.1 V; note 2	–	65	–	ms
		fast start-up; DST = 1; V_{SLPE} steps from 3.1 V to 5.9 V; note 2	–	0.5	–	ms
		fast start-up; DST = 1; V_{SLPE} steps from 5.9 V to 3.1 V; note 2	–	1	–	ms

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Supplies: V _{DD} , V _{MC} , V _P and SLPE						
V _{DD}	operating supply voltage	note 3	2.5	–	6	V
		relaxed performance; note 4	1.8	–	2.5	V
V _{DD} : SUPPLY PIN						
I _{DD}	internal current consumption	V _{DD} = 2.5 V	–	2.3	–	mA
		power-down; PDx = 01; SCL = 1; SDA = 1	–	30	100	μA
		standby; PDx = 11; SCL = 1; SDA = 1	–	2	5	μA
V _{DD} : PERIPHERAL SUPPLY						
I _p	current available for peripheral circuitry	V _{DD} = 2.9 V; RM = 1; SM = 1	–	4.9	–	mA
		V _{DD} = 2.5 V; RM = 1; SM = 1	–	6.5	–	mA
V _{MC} : SENSE INPUT MICROCONTROLLER SUPPLY VOLTAGE						
I _{VMC}	input current	V _{VMC} = 2.5 V	–	4	10	μA
		power-down; PDx = 01; V _{VMC} = 2.5 V; SCL = 1; SDA = 1	–	4	10	μA
		standby; PDx = 11; V _{VMC} = 2.5 V; SCL = 1; SDA = 1	–	2	5	μA
V _P : SUPPLY OUTPUT FOR ELECTRET MICROPHONE						
V _P	output voltage	I _{VP} = 500 μA	1.6	1.9	–	V
Z _{VP}	output impedance	f = 300 Hz	–	40	–	Ω
PSR _{VP}	power supply rejection	f = 300 Hz; note 5	–	65	–	dB
Reset functions: V _{DD} , V _{MC} and RMC						
INTERNAL RESET						
V _{DD(SW)}	switching level of V _{DD} below which internal reset is active	T _{amb} = –10 to +60 °C; note 6	1.0	1.2	1.4	V
RMC: RESET OUTPUT FOR MICROCONTROLLER						
V _{VMC(SW)}	voltage level at pin VMC where RMC changes state	note 7	1.8	2.0	2.2	V
		power-down; PDx = 01; note 7	1.8	2.0	2.2	V
		standby; PDx = 11; note 7	1.8	2.1	2.4	V
ΔV _{VMC} /ΔT	voltage variation with ambient temperature	T _{amb} = –10 to +60 °C	–	0	–	mV/°C
		power-down; PDx = 01; T _{amb} = –10 to +60 °C	–	0	–	mV/°C
		standby; PDx = 11; T _{amb} = –10 to +60 °C	–	+3	–	mV/°C

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Sending channel: MIC+, MIC−, DTMF, OMIC, LN, SCR, REG and LSI						
MIC+ AND MIC−: MICROPHONE INPUTS						
Z _{MIC}	input impedance	differential	60	100	–	kΩ
		single-ended	30	50	–	kΩ
CMRR _{MIC}	common mode rejection ratio	note 8	–	72	–	dB
V _{MIC(peak)}	allowed input signal voltage level (peak value)		–	–	70	mV
G _M	gain MIC+/MIC− to LN		39.5	41	42.5	dB
G _{M(min)}	minimum selectable gain	GMAx = 000100	28.5	30	31.5	dB
G _{M(max)}	maximum selectable gain	GMAx = 011001	49.5	51	52.5	dB
G _{M(step)}	step resolution		–	1	–	dB
ΔG _M	gain variation with frequency	at f = 300 Hz and 3400 Hz with respect to 1 kHz; note 9	–	–	+0.3/−0.7	dB
	gain variation with ambient temperature	at T _{amb} = −10 to +60 °C with respect to 25 °C	–	±0.2	–	dB
	gain variation with line current	at I _{line} = 100 mA with respect to 20 mA; note 9	–	0	±0.5	dB
t _{ACM}	AC start-up time	C _{VDD} = 470 μF; note 10	–	150	–	ms
Sending mute/privacy switch						
ΔG _M	reduction of G _M	SM = 1	–	100	–	dB
DTMF: DUAL TONE MULTI-FREQUENCY INPUT						
R _{DTMF}	parallel input resistance	SM = 1	100	200	–	kΩ
C _{DTMF}	parallel input capacitance	SM = 1	–	45	–	pF
G _{DTMF}	gain from DTMF to LN	SM = 1	20	21	22	dB
G _{DTMF(min)}	minimum selectable gain	SM = 1; GMAx = 100101	0	1	2	dB
G _{DTMF(max)}	maximum selectable gain	SM = 1; GMAx = 001111	20	21	22	dB
G _{DTMF(step)}	step resolution	SM = 1	–	1	–	dB
ΔG _{DTMF}	gain variation with frequency	SM = 1; at f = 300 Hz and 3400 Hz with respect to 1 kHz; note 9	–	–	+0.3/−0.7	dB
	gain variation with ambient temperature	SM = 1; at T _{amb} = −10 to +60 °C with respect to 25 °C	–	±0.2	–	dB
	gain variation with line current	SM = 1; at I _{line} = 100 mA with respect to 20 mA; note 9	–	0	±0.5	dB
Confidence tone						
G _{CTS}	gain from DTMF to QR+/QR−;	RM = 1; SM = 1; notes 11 and 12	–	−25	–	dB
G _{CTS(min)}	minimum selectable gain	RM = 1; SM = 1; GRAx = 111001	–	−40	–	dB
G _{CTS(max)}	maximum selectable gain	RM = 1; SM = 1; GRAx = 100000	–	−19	–	dB
G _{CTS(step)}	step resolution	RM = 1; SM = 1	–	0.5 to 1	–	dB

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
OMIC: MICROPHONE PREAMPLIFIER OUTPUT						
Z _{OMIC}	output impedance		–	400	–	Ω
G _{OMIC}	gain from MIC+/MIC– to OMIC	dynamic limiter not active; note 13	–	20	–	dB
LN: SENDING CHANNEL OUTPUT; notes 14 and 15						
BRL	balance return loss Z _{LN} with Z _{ref} = 220 Ω + (820 Ω // 115 nF)	Z _{line} = ∞ Ω; f = 300 Hz	20	37	–	dB
		Z _{line} = ∞ Ω; f = 1 kHz	20	35	–	dB
		Z _{line} = ∞ Ω; f = 3.4 kHz	20	25	–	dB
Selectable values for Z _{set} = R _a + (R _b // C) with C = 1/(2 π × R _b × f _p); note 16						
R _a	non-shunted resistance of Z _{set}		–	200	–	Ω
R _{a(min)}	minimum selectable value for R _a	ZSAX = 001; note 17	–	0	–	Ω
R _{a(max)}	maximum selectable value for R _a	ZSAX = 11x	–	600	–	Ω
R _{a(step)}	step resolution for R _a		–	100	–	Ω
R _b	shunted resistance of Z _{set}		–	800	–	Ω
R _{b(min)}	minimum selectable value for R _b	ZSBx = 001; notes 17 and 18	–	600	–	Ω
R _{b(max)}	maximum selectable value for R _b	ZSBx = 1x1	–	1000	–	Ω
R _{b(step)}	step resolution for R _b		–	100	–	Ω
f _p	pole frequency determining shunt capacitance C		–	1915	–	Hz
f _{p(min)}	minimum selectable f _p	ZSPx = 0000	–	828	–	Hz
f _{p(max)}	maximum selectable f _p	ZSPx = 0111; note 19	–	5859	–	Hz
n	multiplication factor for f _p	f _p (x + 1) = n × [f _p (x)]	–	1.322	–	
V _{LN(noise)}	noise output voltage	psophometrically weighted (O41 curve)	–	–76	–	dBmp
Dynamic limiter						
V _{LN(p-p)}	threshold of dynamic limiter (peak-to-peak)		3.1	3.5	3.9	V
		DLT = 1	2.2	2.6	3.0	V
		low voltage condition; V _{SLPE} = 3.1 V	–	2.4	–	V
		low current condition; I _{line} = 9 mA	–	2.6	–	V
THD	total harmonic distortion	V _i = 12 mV (RMS) + 10 dB	–	2.5	5.0	%
Dynamic behaviour of limiter; note 20						
t _{att}	attack time	V _i steps from 12 to 38 mV (RMS)	–	1.5	–	ms
t _{rel}	release time	V _i steps from 38 to 12 mV (RMS)	–	90	–	ms

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
SCR: PIN FOR SENDING CURRENT RESISTOR						
V _{SCR}	voltage at pin SCR		–	0.28	–	V
		reduced sending gain; G _M = 30 dB; GMA _x = 000100	–	0.26	–	V
		I _{line} = 12 mA	–	0.22	–	V
		I _{line} = 7 mA	–	0.13	–	V
Receiving channel: LN, LSI, OREC, QR+ and QR–						
QR+, QR–: RECEIVING AMPLIFIER OUTPUTS						
Z _{QR+} , Z _{QR-}	output impedance	single-ended	–	4	–	Ω
G _{RS}	gain from LN to QR+/QR-	note 21	–7.5	-6	–4.5	dB
G _{RS(min)}	minimum selectable gain	GRA _x = 110011	–20.5	–19	–17.5	dB
G _{RS(max)}	maximum selectable gain	GRA _x = 001011	9.5	11.0	12.5	dB
G _{RS(step)}	gain step resolution		–	1	–	dB
ΔG _{RS}	gain variation with frequency	at f = 300 Hz and 3400 Hz with respect to 1 kHz; note 9	–	–	±0.5	dB
ΔG _{RS}	gain variation with temperature	at T _{amb} = –10 to +60 °C with respect to 25 °C	–	±0.2	–	dB
ΔG _{RS}	gain variation with line current	at I _{line} = 100 mA with respect to 20 mA; note 9	–	0	±0.5	dB
t _{ACR}	AC start-up time	C _{VDD} = 470 μF; note 10	–	140	–	ms
V _{QR(p-p)}	maximum output voltage swing (peak-to-peak)	V _{DD} = 5 V; GRA _x = 001011; R _t = ∞ Ω; RFC = 1; V _{LN} = 2 V (RMS)	–	2.3	–	V
		HPL = 1; V _{DD} = 5 V; GRA _x = 001011; R _t = ∞ Ω; RFC = 1; V _{LN} = 2 V (RMS)	–	5.9	–	V
V _{QR(rms)}	output voltage (RMS value); THD = 2%	HPL = 1; GRA _x = 000011; note 22	0.45	–	–	V
		HPL = 1; R _t = 450 Ω; GRA _x = 000011; note 22	0.84	–	–	V
		RFC = 1; C _t = 80 nF; f = 3.4 kHz; GRA _x = 000011; note 22	0.9	–	–	V
		single-ended; HPL = 1; Z _t = 150 Ω + 100 μF at QR–; GRA _x = 001001; note 22	0.45	–	–	V
V _{QR(noise)}	noise output voltage	psophometrically weighted (O41 curve)	–	–82	–	dBmp
V _{QR(offset)}	DC offset voltage between QR+/QR–		–	–	±100	mV
OREC: OUTPUT RECEIVE PREAMPLIFIER						
Z _{OREC}	output impedance		–	1000	–	Ω
G _{OREC}	gain from LN to OREC	note 13	–	–6	–	dB

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Selectable values for $Z_{oss} = R_{sa} (R_{sb} // C_s)$; note 23						
R _{sa}	non-shunted resistance of Z _{oss}		–	492	–	Ω
R _{sa(min)}	minimum selectable value R _{sa}	ZOSAx = 0000	–	134	–	Ω
R _{sa(max)}	maximum selectable value for R _{sa}	ZOSAx = 1010; note 24	–	492	–	Ω
R _{sb}	shunted resistance of Z _{oss}		–	1259	–	Ω
R _{sb(min)}	minimum selectable value for R _{sb}	ZOSBx = 0000	–	465	–	Ω
R _{sb(max)}	maximum selectable value for R _{sb}	ZOSBx = 1011; note 24	–	2216	–	Ω
C _s	shunt capacitance of Z _{oss}		–	134	–	nF
C _{s(min)}	minimum selectable value for C _s	ZOSPx = 0001; note 25	–	55	–	nF
C _{s(max)}	maximum selectable value for C _s	ZOSPx = 1111; note 24	–	259	–	nF
Sidetone suppression; note 26						
G _{STS}	gain from MIC+/MIC– to QR+/QR–	Z _{line} = 492 Ω + (1259 Ω // 134 nF); f = 300 Hz	–	11	15	dB
		Z _{line} = 492 Ω + (1259 Ω // 134 nF); f = 1 kHz	–	5	10	dB
		Z _{line} = 492 Ω + (1259 Ω // 134 nF); f = 3.4 kHz	–	9	15	dB
Dial output connection: DOC (open-drain output)						
I _{DOC}	output sink current	V _{DOC} = 12 V	–	0	100	nA
		DPI = 1; V _{DOC} = 0.4 V; V _{DD} = 2.5 V	200	400	–	μA
Line current control: LN and SLPE						
I _{line(min)}	minimum value of DC line current that can be read as a bit code via the I ² C-bus	LCx = 00001	–	15	–	mA
I _{line(max)}	maximum value of DC line current that can be read as a bit code via the I ² C-bus	LCx = 11110	–	91	–	mA
I _{line(step)}	DC line current step resolution	note 27	–	≈2.5	–	mA
I ² C-bus inputs/outputs: SDA and SCL						
	in accordance with standard	note 28	–	–	–	

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Clock input: CLK						
$V_{CLK(p-p)}$	input signal voltage level (peak-to-peak value)		200	–	$V_{VMC} - V_{SS}$	mV
$\Delta f_{CLK}/f_{CLK}$	frequency tolerance	note 29	–	–	± 0.5	%
R_{CLK}	input series resistance		–	800	–	k Ω
C_{CLK}	input series capacitance		–	4	–	pF

Notes

- Time needed to reach at start-up the default DC voltage V_{SLPE} ($\pm 10\%$ from its final value):
 - Time depends strongly on the value of the capacitor(s) at V_{DD} and VMC ; with a lower value of C_{VDD} the DC start-up time decreases.
 - The start-up time can be reduced considerably by programming the bit code $DST = 1$ during the start-up procedure. In practice this is possible as soon as the microcontroller has become operational.
- Time needed to reach the DC voltage V_{SLPE} within $\pm 10\%$ from its final value) after reprogramming $VDCx$.
- The supply voltage V_{DD} is determined by the regulated DC voltage at pin $SLPE$ and by the voltage drop between pin $SLPE$ and V_{DD} ; see Chapter "Functional description".
- Relaxed performance means: parameters can deviate from their specified values.
- Rejection between supply pin V_{DD} and V_P . Rejection between pin LN and V_P can be calculated by adding the attenuation of the first-order low-pass filter ($R = 250\ \Omega$, $C = 150\ \mu F$) between $SLPE$ and V_{DD} .
- If V_{DD} is above this level, the default values have been loaded into the internal registers.
- RMC changes from logic 1 to logic 0 when voltage on pin VMC is increasing; RMC changes from logic 0 to logic 1 when voltage on pin VMC is decreasing; see Fig.4.
- Common mode signal is applied via $2 \times 470\ \Omega$ external resistors connected to pins $MIC+$ and $MIC-$.
- Not tested, guaranteed by design.
- Time needed to reach default settings ($\pm 3\ dB$).
- At low gain settings the confidence tone gain will be slightly higher than the specified value due to a residual signal.
- G_{CTA} , the confidence tone gain for asymmetrical drive, equals $G_{CTS} - 6$ (in dB).
- To be left open-circuit in application.
- The AC set impedance between pin LN and V_{SS} consists of $R_a + (R_b // C)$ in parallel with an artificial inductor L_{eq} and internal resistors R_p and R_{LSI} and internal capacitor C_p . See Chapter "Functional description".
- Balance Return Loss indicates the deviation of an impedance with respect to a reference impedance.
 $BRL = 20 \log |(Z_{LN} + Z_{ref})/(Z_{LN} - Z_{ref})|$ where $Z_{LN} \approx R_a + (R_b // C)$ is the impedance seen into pin LN
 $Z_{ref} = R_{a(ref)} + (R_{b(ref)} // C_{ref})$ is the reference impedance.
- Without clock the set impedance is automatically set to $Z_{set} = 600\ \Omega$ (typical).
- The combination $R_a = 0$ and $R_b = 0$ is not allowed (see Tables 9 and 10, note 1).
- Value logic 0 can also be programmed.
- Value $f_p = 12\ kHz$ can also be programmed.
- Attack and release times are also valid under low current and voltage conditions.
- G_{RA} , the receiving channel gain for asymmetrical drive equals $G_{RS} - 6$ (in dB).
- The maximum possible output swing depends on the DC conditions (the programmed voltage V_{SLPE} and the load on the supply pin V_{DD}) and on the gain setting of the receiving channel.

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23. The internal balance impedance Z_{oss} to match the external load impedance at pin LN ($Z_{line} = Z_{oss}$) for optimum sidetone suppression; $Z_{oss} = R_{sa} + (R_{sb} // C_s)$; without clock the sidetone balance impedance is automatically set to $Z_{oss} = 600 \Omega$ (typical).
24. Other values can be found in Tables 14, 15 and 16.
25. Value $C_s = 5 \text{ nF}$ can also be programmed.
26. Gain sending channel $G_M = \text{default}$ (typical 41 dB); gain receiving channel $G_{rec} = \text{default}$ (typical -6 dB); sidetone gain G_{STS} minimum sidetone suppression at $f = 300 \text{ Hz}$ and 3400 Hz is: $G_M + G_R - G_{st(max)} = 41 - 6 - 15 = 20 \text{ dB}$. G_{STA} , the sidetone gain for asymmetrical drive equals $G_{STS} - 6$ (in dB).
27. Indication only; exact values can be found in Table 16.
28. Standard I²C-bus specifications are valid for $V_{DD} \geq 2.5 \text{ V}$. Relaxed specifications for $V_{DD} = 1.8$ to 2.5 V .
29. Recommended accuracy of input frequency; a higher tolerance will cause parameters to deviate from their specified values; note that all parameters are specified with the reference input clock frequency $f_{clk} = 3.579545 \text{ MHz}$.

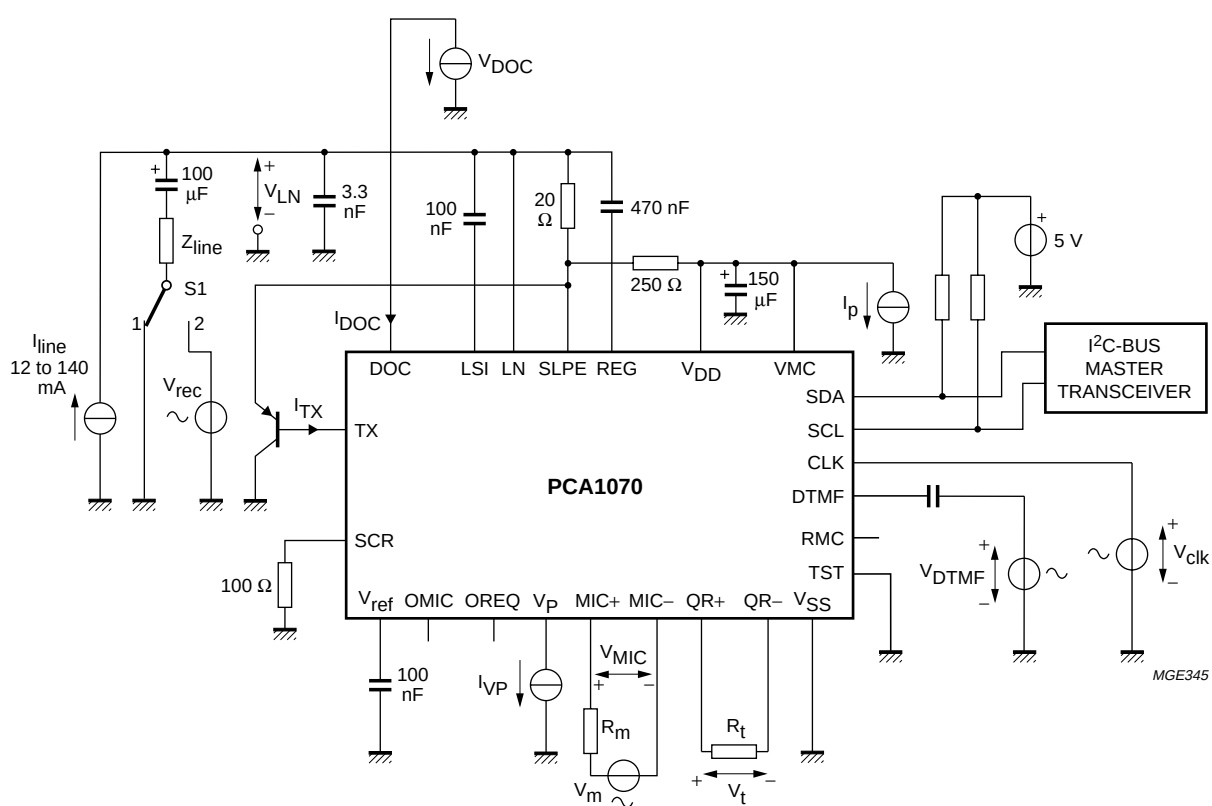
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TEST AND APPLICATION INFORMATION

The test circuit is illustrated in Fig.7. The basic application circuit is illustrated in Fig.8. An interrupter with an N-channel depletion MOS transistor (e.g. BSD254A or BSP124) is shown. It is intended for applications where a low DC line voltage is required. An interrupter with an N-channel enhancement MOS transistor (e.g. BSN304A or BSP130) can be used for applications where a relatively high DC line voltage is allowed.

An application circuit for applications where a low DC line voltage and long line interrupts are required, is illustrated in Fig.9 (interrupter with an N-channel depletion MOS transistor).



Definitions:

Gain sending channel $G_M = 20 \log (V_{LN}/V_{MIC})$ with S1 in position 1; $V_{DTMF} = 0$.

Gain DTMF amplifier $G_{\text{DTMF}} = 20 \log (V_{\text{LN}}/V_{\text{DTMF}})$ with S1 in position 1; $V_m = 0$.

Gain receiving channel $G_{\text{rec}} = 20 \log (V_T/V_N)$ with S1 in position 2; $V_m = 0$.

Sidetone gain $G_{ST} = 20 \log (V_T/V_{MIC})$ with S1 in position 1; $V_{DTMF} = 0$.

Fig.7 Test circuit of the PCA1070.

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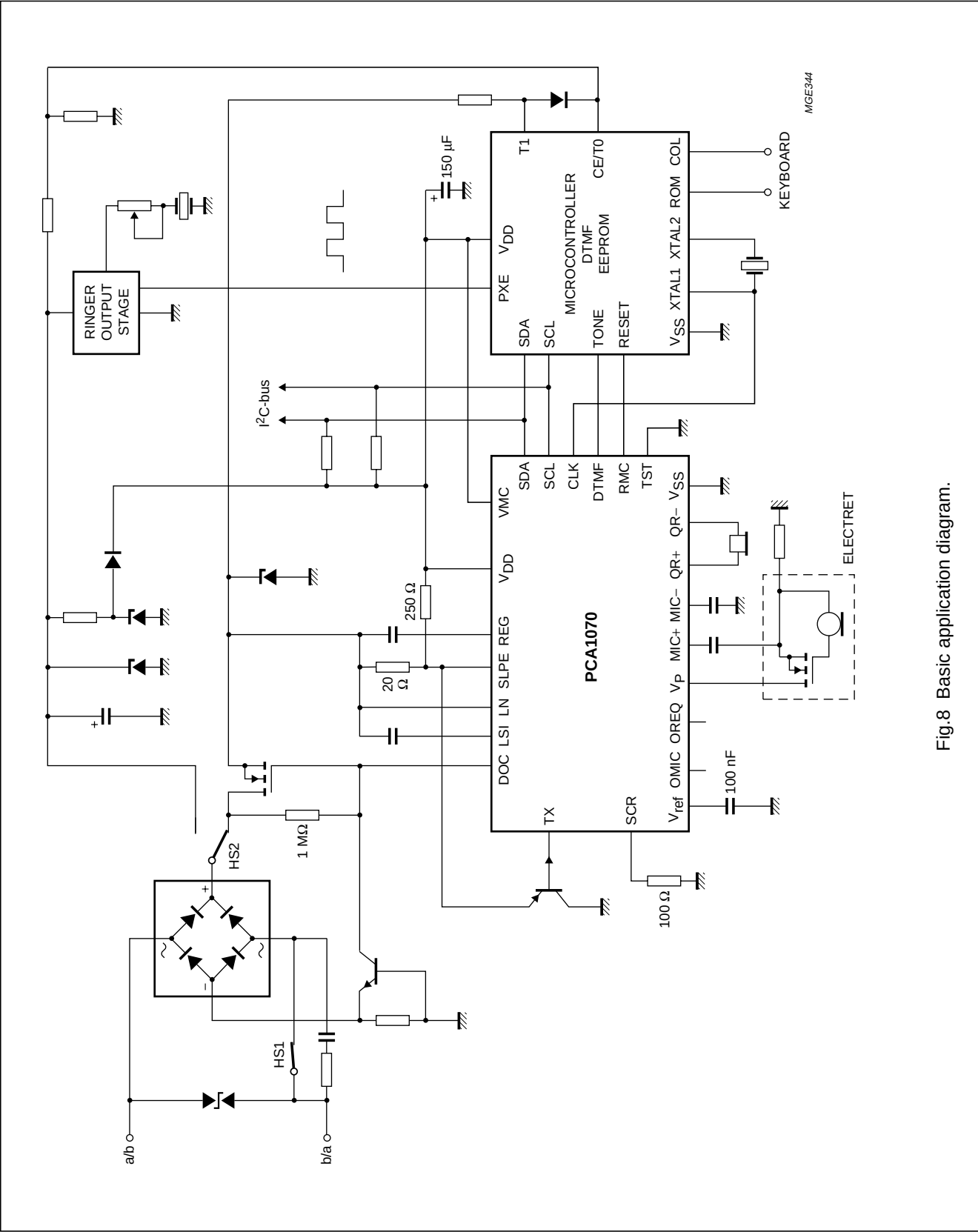


Fig.8 Basic application diagram.

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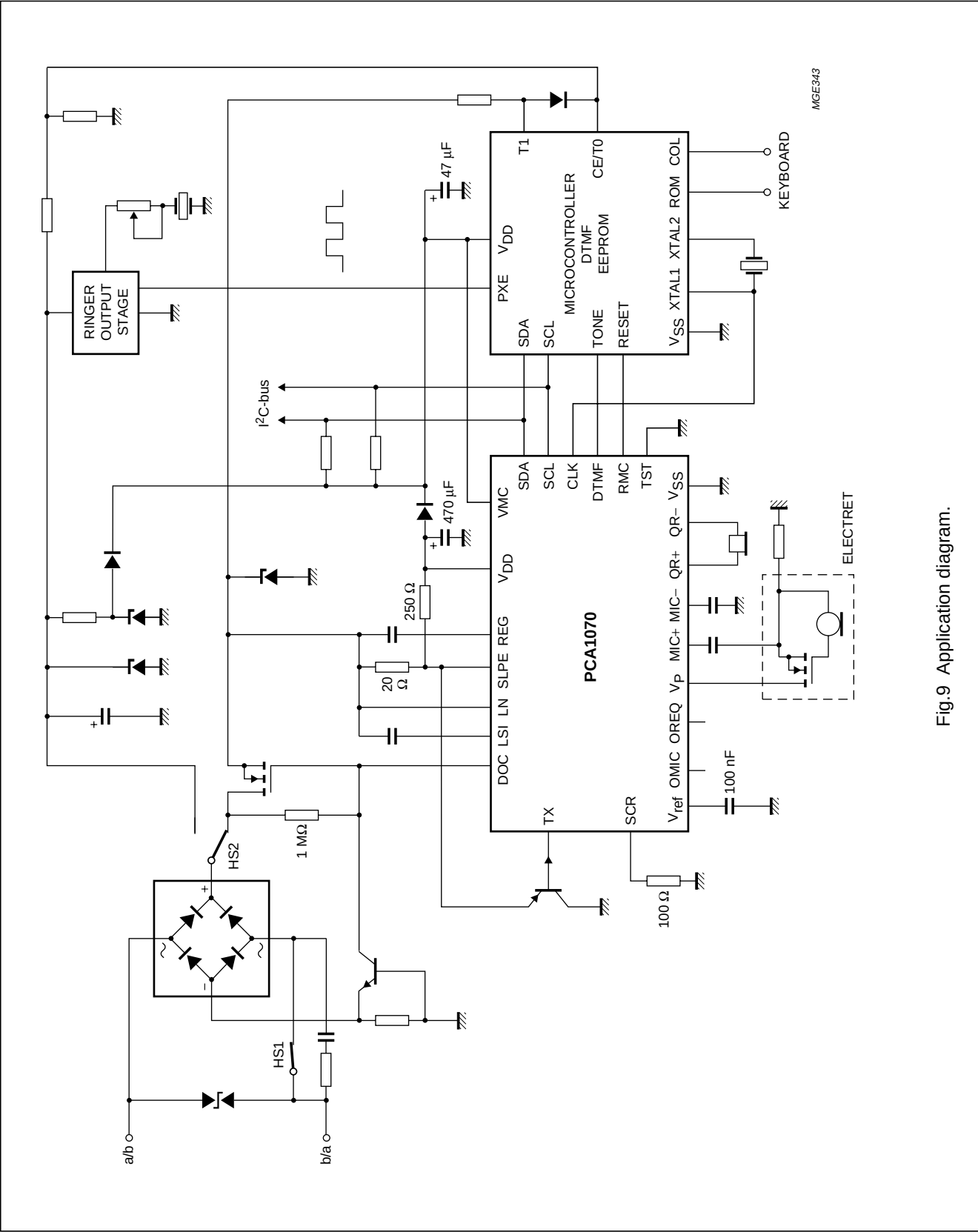


Fig.9 Application diagram.

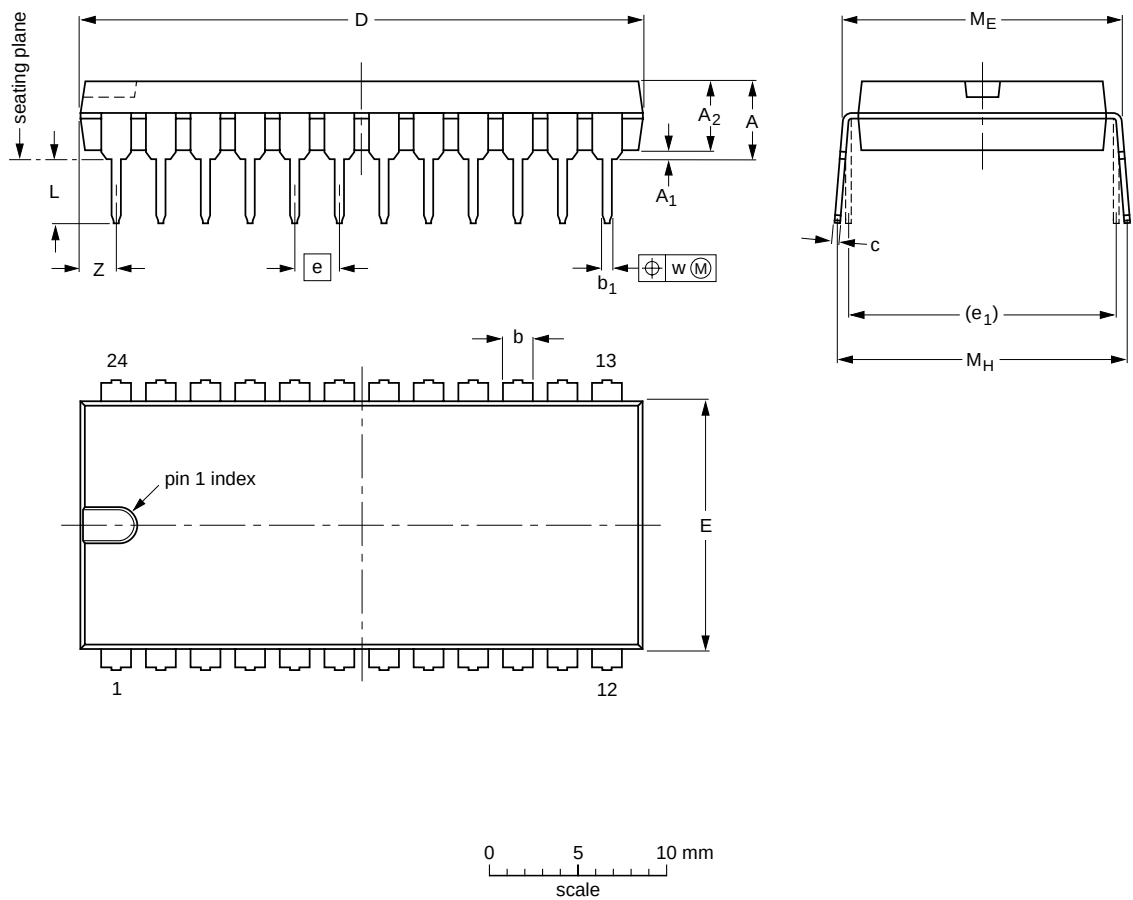
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PACKAGE OUTLINES

DIP24: plastic dual in-line package; 24 leads (600 mil)

SOT101-1



DIMENSIONS (inch dimensions are derived from the original mm dimensions)

UNIT	A max.	A ₁ min.	A ₂ max.	b	b ₁	c	D ⁽¹⁾	E ⁽¹⁾	e	e ₁	L	M _E	M _H	w	Z ⁽¹⁾ max.
mm	5.1	0.51	4.0	1.7 1.3	0.53 0.38	0.32 0.23	32.0 31.4	14.1 13.7	2.54	15.24	3.9 3.4	15.80 15.24	17.15 15.90	0.25	2.2
inches	0.20	0.020	0.16	0.066 0.051	0.021 0.015	0.013 0.009	1.26 1.24	0.56 0.54	0.10	0.60	0.15 0.13	0.62 0.60	0.68 0.63	0.01	0.087

Note

1. Plastic or metal protrusions of 0.25 mm maximum per side are not included.

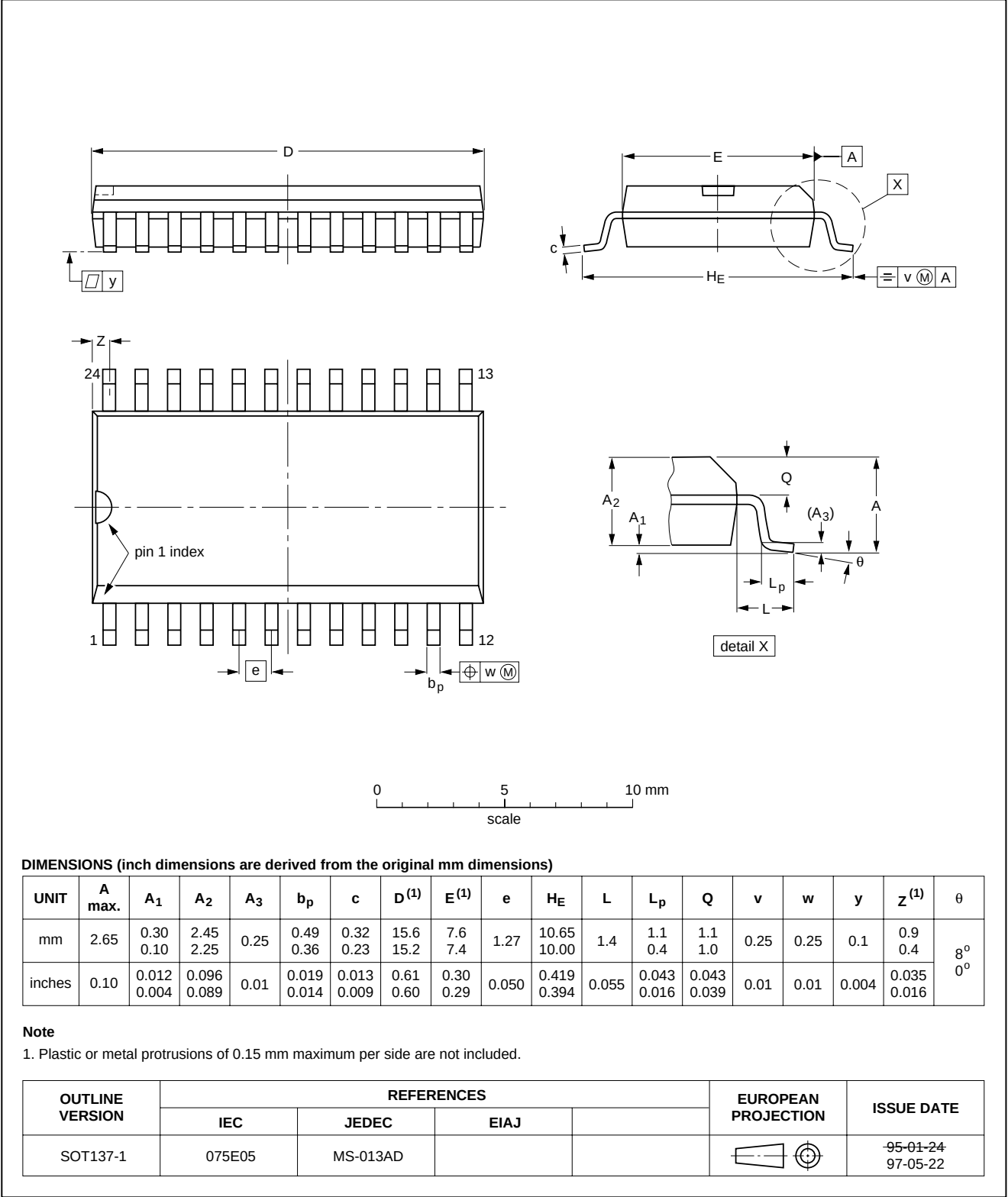
OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT101-1	051G02	MO-015AD				92-11-17 95-01-23

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SO24: plastic small outline package; 24 leads; body width 7.5 mm

SOT137-1



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SOLDERING

Introduction

There is no soldering method that is ideal for all IC packages. Wave soldering is often preferred when through-hole and surface mounted components are mixed on one printed-circuit board. However, wave soldering is not always suitable for surface mounted ICs, or for printed-circuits with high population densities. In these situations reflow soldering is often used.

This text gives a very brief insight to a complex technology. A more in-depth account of soldering ICs can be found in our *"IC Package Databook"* (order code 9398 652 90011).

DIP

SOLDERING BY DIPPING OR BY WAVE

The maximum permissible temperature of the solder is 260 °C; solder at this temperature must not be in contact with the joint for more than 5 seconds. The total contact time of successive solder waves must not exceed 5 seconds.

The device may be mounted up to the seating plane, but the temperature of the plastic body must not exceed the specified maximum storage temperature ($T_{stg\ max}$). If the printed-circuit board has been pre-heated, forced cooling may be necessary immediately after soldering to keep the temperature within the permissible limit.

REPAIRING SOLDERED JOINTS

Apply a low voltage soldering iron (less than 24 V) to the lead(s) of the package, below the seating plane or not more than 2 mm above it. If the temperature of the soldering iron bit is less than 300 °C it may remain in contact for up to 10 seconds. If the bit temperature is between 300 and 400 °C, contact may be up to 5 seconds.

SO

REFLOW SOLDERING

Reflow soldering techniques are suitable for all SO packages.

Reflow soldering requires solder paste (a suspension of fine solder particles, flux and binding agent) to be applied to the printed-circuit board by screen printing, stencilling or pressure-syringe dispensing before package placement.

Several techniques exist for reflowing; for example, thermal conduction by heated belt. Dwell times vary between 50 and 300 seconds depending on heating method. Typical reflow temperatures range from 215 to 250 °C.

Preheating is necessary to dry the paste and evaporate the binding agent. Preheating duration: 45 minutes at 45 °C.

WAVE SOLDERING

Wave soldering techniques can be used for all SO packages if the following conditions are observed:

- A double-wave (a turbulent wave with high upward pressure followed by a smooth laminar wave) soldering technique should be used
- The longitudinal axis of the package footprint must be parallel to the solder flow
- The package footprint must incorporate solder thieves at the downstream end.

During placement and before soldering, the package must be fixed with a droplet of adhesive. The adhesive can be applied by screen printing, pin transfer or syringe dispensing. The package can be soldered after the adhesive is cured.

Maximum permissible solder temperature is 260 °C, and maximum duration of package immersion in solder is 10 seconds, if cooled to less than 150 °C within 6 seconds. Typical dwell time is 4 seconds at 250 °C.

A mildly-activated flux will eliminate the need for removal of corrosive residues in most applications.

REPAIRING SOLDERED JOINTS

Fix the component by first soldering two diagonally-opposite end leads. Use only a low voltage soldering iron (less than 24 V) applied to the flat part of the lead. Contact time must be limited to 10 seconds at up to 300 °C. When using a dedicated tool, all other leads can be soldered in one operation within 2 to 5 seconds between 270 and 320 °C.

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DEFINITIONS

Data sheet status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of this specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

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NOTES

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NOTES

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Printed in The Netherlands

417027/00/03/pp36

Date of release: 1997 Jun 20

Document order number: 9397 750 00949

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